

Numerical Rank Certification of Interpolation Moment Matrices by Column-Pivoted Householder QR Factorization

Abstract

The construction of generalized interpolation weights on scattered three-dimensional stencils requires the solution of a dense 27×27 moment system $\mathbf{M}\Psi = \mathbf{C}$ with $\mathbf{M} = \mathbf{X}^\top \mathbf{W}$, where $\mathbf{X}$ collects polynomial samples up to degree six and $\mathbf{W}$ collects Hermite–Wendland radial basis rows. Admissibility of a candidate stencil–parameter triple (stencil, β, h) hinges on whether $\mathbf{M}$ possesses full numerical rank in IEEE double precision. This note places the rank test used in our solver on a rigorous and fully referenced footing. We first review the Householder reflection and its role in orthogonal triangularization (Section 1), then the QR factorization with column pivoting (QRCP) of Businger and Golub and its rank-revealing interpretation (Section 2), and finally state the complete certification algorithm employed in our code: iterative row/column max-norm equilibration, a three-stage sequence of rank gates, and the acceptance criterion $p_i > \tau_{\widetilde{\mathbf{M}}}$ with $\tau_{\widetilde{\mathbf{M}}} = 27 \epsilon_{\text{dbl}} \|\widetilde{\mathbf{M}}\|_F$ (Section 3). The threshold is shown to follow the standard backward-error convention of numerical linear algebra, and the result is characterized precisely as a double-precision numerical certificate rather than a symbolic proof of nonsingularity.

1 Householder reflections and orthogonal triangularization

The elementary orthogonal transformation underlying all algorithms in this note is the Householder reflection, introduced by Householder [1] as a tool for the unitary triangularization of general matrices; systematic treatments are given by Golub and Van Loan [3, §5.1] and Trefethen and Bau [4, Lecture 10].

Definition 1.1 (Householder reflector). Let $\mathbf{v} \in \mathbb{R}^m$, $\mathbf{v} \neq \mathbf{0}$. The matrix

$$\mathbf{H} = \mathbf{I} - 2 \frac{\mathbf{v}\mathbf{v}^\top}{\mathbf{v}^\top \mathbf{v}} \quad (1)$$

is called a Householder reflector with Householder vector $\mathbf{v}$.

Geometrically, $\mathbf{H}$ realizes the reflection of $\mathbb{R}^m$ across the hyperplane $\{\mathbf{x} : \mathbf{v}^\top \mathbf{x} = 0\}$ orthogonal to $\mathbf{v}$ [4, Lecture 10]. Two properties follow directly from (1): $\mathbf{H}$ is symmetric, $\mathbf{H} = \mathbf{H}^\top$, and orthogonal, $\mathbf{H}^\top \mathbf{H} = \mathbf{I}$, hence involutory, $\mathbf{H}^2 = \mathbf{I}$.

The utility of the reflector lies in its ability to annihilate all but the first component of a prescribed vector.

Lemma 1.2 (Annihilation property). Let $\mathbf{a} \in \mathbb{R}^m$ with $\alpha = \|\mathbf{a}\|_2 > 0$ and $\mathbf{a} \neq \alpha \mathbf{e}_1$. Choose $\mathbf{v} = \mathbf{a} - \alpha \mathbf{e}_1$ and let $\mathbf{H}$ be the reflector (1). Then

$$\mathbf{H}\mathbf{a} = \alpha \mathbf{e}_1. \quad (2)$$

Proof. Write $a_1 = \mathbf{e}_1^\top \mathbf{a}$. Direct computation gives

$$\mathbf{v}^\top \mathbf{a} = (\mathbf{a} - \alpha \mathbf{e}_1)^\top \mathbf{a} = \alpha^2 - \alpha a_1, \quad (3)$$

$$\mathbf{v}^\top \mathbf{v} = (\mathbf{a} - \alpha \mathbf{e}_1)^\top (\mathbf{a} - \alpha \mathbf{e}_1) = \alpha^2 - 2\alpha a_1 + \alpha^2 = 2(\alpha^2 - \alpha a_1). \quad (4)$$

Hence $\mathbf{v}^\top \mathbf{a} / \mathbf{v}^\top \mathbf{v} = \frac{1}{2}$, and therefore

$$\mathbf{H}\mathbf{a} = \mathbf{a} - 2 \frac{\mathbf{v}^\top \mathbf{a}}{\mathbf{v}^\top \mathbf{v}} \mathbf{v} = \mathbf{a} - \mathbf{v} = \alpha \mathbf{e}_1. \quad (5)$$

□

In finite-precision implementations the sign of α is chosen as $\alpha = -\text{sign}(a_1)\|\mathbf{a}\|_2$ so that the subtraction forming $\mathbf{v}$ incurs no cancellation; see Higham [5, §19.1] and [3, §5.1.3].

Applying Lemma 1.2 recursively to the trailing submatrices of $\mathbf{A} \in \mathbb{R}^{m \times n}$ produces the Householder QR factorization: after n steps,

$$\mathbf{H}_n \cdots \mathbf{H}_2 \mathbf{H}_1 \mathbf{A} = \mathbf{R}, \quad \mathbf{A} = \mathbf{Q}\mathbf{R}, \quad \mathbf{Q} = \mathbf{H}_1 \mathbf{H}_2 \cdots \mathbf{H}_n, \quad (6)$$

with $\mathbf{R}$ upper triangular and $\mathbf{Q}$ orthogonal. The decisive advantage of this construction over Gram–Schmidt or Gaussian elimination is its backward stability: the computed factors are the exact factors of a perturbed matrix $\mathbf{A} + \Delta\mathbf{A}$ with $\|\Delta\mathbf{A}\|_F \leq c(m, n) \epsilon_{\text{dbl}} \|\mathbf{A}\|_F$ for a modest polynomial $c(m, n)$, where $\epsilon_{\text{dbl}} \approx 2.22 \times 10^{-16}$ denotes the unit roundoff of IEEE binary64 arithmetic; see Higham [5, Thm. 19.4] and [3, §5.1.9]. This backward-error bound is the quantitative basis of the rank threshold adopted in Section 3.

2 QR factorization with column pivoting and numerical rank

2.1 The Businger–Golub algorithm

The QR factorization with column pivoting (QRCP) was introduced by Businger and Golub [2] in the context of linear least-squares problems with (nearly) rank-deficient coefficient matrices; the canonical modern description is [3, §5.4.2]. Given $\mathbf{A} \in \mathbb{R}^{m \times n}$, QRCP computes a permutation matrix $\mathbf{P}$, an orthogonal $\mathbf{Q}$, and an upper-triangular $\mathbf{R}$ such that

$$\mathbf{A}\mathbf{P} = \mathbf{Q}\mathbf{R}. \quad (7)$$

At step k the algorithm selects, among the not-yet-processed columns, the one whose trailing subcolumn (rows $k:m$) has largest Euclidean norm, swaps it into position k , and annihilates its subdiagonal entries with a Householder reflector (Lemma 1.2). The greedy pivot selection enforces the monotonicity

$$|r_{11}| \geq |r_{22}| \geq \cdots \geq |r_{qq}|, \quad q = \min(m, n), \quad (8)$$

and, more strongly, guarantees that each diagonal entry dominates the trailing block column-wise, $r_{kk}^2 \geq \sum_{i=k}^j r_{ij}^2$ for all $j \geq k$ [3, §5.4.2]. The reference implementation in LAPACK is the routine `xGEQP3` [10], the BLAS-3 formulation of Quintana-Ortí, Sun, and Bischof [9]. The algorithm is deterministic: the pivot at every step is defined by exact column-norm comparisons, so repeated executions on the same data produce identical factorizations, in contrast to randomized rank estimators.

2.2 Mechanics of the pivot residual: trailing norms and their evolution

The quantity compared against the rank tolerance in Section 3 is the *pivot residual* p_k of step k . This subsection makes its definition, its evolution under the reflections, and its geometric meaning precise, since all three are essential to a correct reading of the acceptance test: the residual is a property of the *partially factored* working matrix, not of the raw data.

Let $\mathbf{A}^{(0)} = \mathbf{A} \in \mathbb{R}^{m \times n}$ denote the (preprocessed) input and let

$$\mathbf{A}^{(k)} = \mathbf{H}_{k-1} \cdots \mathbf{H}_1 \mathbf{H}_0 \mathbf{A} \mathbf{P}_0 \mathbf{P}_1 \cdots \mathbf{P}_{k-1} \quad (9)$$

be the working matrix after k pivoted reflection steps. For every not-yet-selected column j , define its *trailing norm* at step k as the Euclidean norm of its subvector below the already triangularized block,

$$\nu_k(j) = \left(\sum_{i=k}^{m-1} (a_{ij}^{(k)})^2 \right)^{1/2}, \quad p_k = \max_{j \text{ unselected}} \nu_k(j). \quad (10)$$

The leading components $a_{ij}^{(k)}$, $i < k$, are excluded: after the first k reflections they carry the coefficients of column j with respect to the already selected directions—entries of rows $0, \dots, k-1$ of $\mathbf{R}$ —and therefore contribute nothing new to linear independence. At $k=0$ the trailing norm coincides with the raw column norm; for every $k \geq 1$ it is a quantity *manufactured by the preceding reflections*.

Proposition 2.1 (Pivot residual equals the produced diagonal entry). *At step k , let column j^* attain the maximum in (10) and be swapped into position k . Then the diagonal entry created by the k th Householder reflection satisfies*

$$|r_{kk}| = \nu_k(j^*) = p_k. \quad (11)$$

Consequently the acceptance test $p_k > \tau$ may be evaluated before the reflection is formed, with a result identical to inspecting $|r_{kk}|$ afterwards.

Proof. Write $\mathbf{x} \in \mathbb{R}^{m-k}$ for the trailing subvector of the pivot column, so $\|\mathbf{x}\|_2 = \nu_k(j^*)$. By Lemma 1.2 the reflector acting on components $k, \dots, m-1$ maps $\mathbf{x}$ to $\alpha \mathbf{e}_1$ with $|\alpha| = \|\mathbf{x}\|_2$, and α is precisely the entry stored in position (k, k) of $\mathbf{R}$. Orthogonality of the reflector preserves the Euclidean length, so no other value is possible. $\square$

Geometrically, the leading/trailing split of a working column is its orthogonal decomposition with respect to the span of the selected pivot columns:

$$\underbrace{(a_{0j}^{(k)}, \dots, a_{k-1,j}^{(k)})}_{\text{component inside the selected subspace}} \quad \Bigg| \quad \underbrace{(a_{kj}^{(k)}, \dots, a_{m-1,j}^{(k)})}_{\text{component orthogonal to it}}, \quad (12)$$

so $\nu_k(j)$ is the Euclidean distance from column j to the subspace spanned by the k columns already selected. A large residual certifies a genuinely new direction; a residual below the roundoff-level tolerance means the column lies, to double precision, inside the subspace already captured—numerical linear dependence. This is why the pivot residual measures *independence* rather than mere *magnitude*, and why raw column norms cannot replace it, as the following example shows.

Example 2.2 (Raw norms cannot reveal rank). Let $\delta = 10^{-15}$ and consider the two columns

$$\mathbf{a}_1 = (1, 0, 0)^\top, \quad \mathbf{a}_2 = (1, \delta, 0)^\top, \quad (13)$$

which are numerically parallel. Their raw norms are $\|\mathbf{a}_1\|_2 = 1$ and $\|\mathbf{a}_2\|_2 = \sqrt{1 + \delta^2} \approx 1$; a test based on raw norms would count two directions and report rank 2. Under QRCP, step 0 selects $\mathbf{a}_1$ ($p_0 = 1$), and the reflection moves the component of $\mathbf{a}_2$ along $\mathbf{a}_1$ into the leading position, leaving the trailing norm $\nu_1(2) = \delta = 10^{-15}$. Against any tolerance of the backward-error magnitude (16) this residual fails, $p_1 \leq \tau$, and the numerical rank is correctly reported as 1. The reflections thus convert magnitude information into independence information; omitting them (or measuring full-length norms of the unreflected data) invalidates the test from the second step onward.

Two further consequences of orthogonal invariance fix the roles of the two norms appearing in the acceptance test $p_k > \tau$.

Proposition 2.3 (The tolerance is frozen; the residuals evolve). *The Frobenius norm is invariant under orthogonal transformations and column permutations, $\|\mathbf{H}\mathbf{A}\mathbf{P}\|_F = \|\mathbf{A}\|_F$, since $\|\mathbf{H}\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}^\top \mathbf{H}^\top \mathbf{H} \mathbf{A}) = \|\mathbf{A}\|_F^2$. Hence the tolerance $\tau = f(m, n) \epsilon_{\text{dbl}} \|\mathbf{A}^{(0)}\|_F$, computed once from the preprocessed input before the factorization begins, retains its value in exact arithmetic throughout all steps; recomputing it would change nothing except adding roundoff. The test $p_k > \tau$ therefore compares an evolving measurement (the trailing norm, updated by every reflection) against a fixed ruler (the roundoff-level noise bound of the whole matrix).*

Remark 2.4 (Interleaved termination). Textbook descriptions sometimes present rank determination as a post-processing pass over the completed diagonal of $\mathbf{R}$. By Proposition 2.1 and the monotonicity (8), the interleaved variant—test $p_k > \tau$ at the top of every step and terminate at the first failure—produces the identical rank count while offering two advantages. First, efficiency: since p_k is the *maximum* remaining residual, its failure guarantees that all subsequent pivots fail as well, so the remaining reflections are provably useless work. Second, numerical hygiene: forming a reflector from a pivot of roundoff-level norm divides by the near-zero quantity $\mathbf{v}^\top \mathbf{v}$ and sprays amplified noise into all remaining trailing blocks; stopping before the degenerate reflection keeps the working matrix uncontaminated. The rank-deficient least-squares driver `xGELSY` of LAPACK adopts the same interleaved strategy [10].

2.3 Rank-revealing interpretation

Because $\mathbf{Q}$ is orthogonal, the singular values of $\mathbf{R}$ coincide with those of $\mathbf{A}$, and the profile of the diagonal (8) exposes the numerical rank: if $|r_{kk}|$ is large for $k \leq r$ and collapses to roundoff level for $k > r$, then $\mathbf{A}$ has numerical rank r . Factorizations of the form (7) whose triangular factor reliably separates the dominant and negligible parts of the spectrum were named *rank-revealing QR* (RRQR) factorizations by Chan [6]. Interlacing arguments give the a priori bounds

$$\sigma_k(\mathbf{A}) \geq |r_{kk}| \quad \text{is not guaranteed in general, but} \quad |r_{kk}| \leq \sqrt{n-k+1} \sigma_k(\mathbf{A}), \quad (14)$$

so a small pivot certifies a small singular value; the converse can fail for contrived matrices. The classical counterexample is the Kahan matrix [8], for which the Businger–Golub pivots decay far more slowly than the smallest singular value. Gu and Eisenstat [7] constructed *strong* RRQR algorithms with provable factor bounds that close this gap at modest extra cost. In practice, however, QRCP fails to reveal rank only on such adversarial examples; for equilibrated matrices of moderate dimension it is the standard, inexpensive alternative to the singular value decomposition, and it underlies the rank-deficient least-squares driver `xGELSY` of LAPACK [10].

2.4 Numerical rank and tolerance conventions

For a matrix stored and factored in floating-point arithmetic, rank is meaningful only relative to a tolerance τ [3, §5.4.1]: one defines

$$\text{rank}_{\text{num}}(\mathbf{A}; \tau) = \#\{k : \sigma_k(\mathbf{A}) > \tau\} \quad \text{or, in the QRCP surrogate,} \quad \#\{k : p_k > \tau\}, \quad (15)$$

where p_k denotes the k th pivot magnitude (the trailing column-norm residual, equal to $|r_{kk}|$ at selection time). The tolerance must dominate the backward error of the computation itself; otherwise the test would attribute significance to quantities that are indistinguishable from roundoff. Since Householder QR applied to $\mathbf{A} \in \mathbb{R}^{n \times n}$ is backward stable with perturbation bound proportional to $n \epsilon_{\text{dbl}} \|\mathbf{A}\|$ [5, Thm. 19.4], tolerances of the generic form

$$\tau = f(m, n) \epsilon_{\text{dbl}} \|\mathbf{A}\| \quad (16)$$

are the accepted convention, with f a low-degree polynomial of the dimensions and $\|\cdot\|$ a computable norm. Instances of this family include the default tolerance $\max(m, n) \epsilon_{\text{dbl}} \sigma_{\max}(\mathbf{A})$ used for numerical rank determination in standard software following [3, §5.4.1], and the relative-pivot condition $|r_{kk}|/|r_{11}| > \text{RCOND}$ of `xGELSY` [10]. The particular member of the family used in our solver is specified and justified next.

3 The certification algorithm

3.1 Setting

For every candidate triple (stencil, β, h) of a 27-point interpolation stencil, Wendland shape parameter β , and support radius h , our generalized-interpolation solver assembles a moment matrix

$$\mathbf{M} = \mathbf{X}^\top \mathbf{W} \in \mathbb{R}^{27 \times 27}, \quad (17)$$

where the columns of $\mathbf{X}$ sample the polynomial basis of total degree at most six on the stencil nodes and the columns of $\mathbf{W}$ contain Hermite–Wendland radial-basis rows [13, 14]. The interpolation weights Ψ solve $\mathbf{M}\Psi = \mathbf{C}$ for a prescribed moment vector $\mathbf{C}$. A triple is *admissible* if and only if this system is uniquely solvable in double precision, i.e. if $\mathbf{M}$ has full numerical rank 27 in the sense of (15). Existence and uniqueness of Ψ is governed by $\text{rank}(\mathbf{M})$ alone; the auxiliary tests on $\mathbf{X}$ and $\mathbf{W}$ described below are diagnostic accelerators, not additional mathematical requirements.

3.2 Row/column equilibration

The entries of $\mathbf{M}$ mix polynomial degrees one through six and can span many orders of magnitude. A pivot-magnitude test applied to the raw matrix could then mistake benign scale imbalance for rank deficiency, since column norms—the very quantities QRCP pivots on—would reflect degree scaling rather than linear independence. Diagonal scaling is the classical remedy: van der Sluis [11] showed that norm equilibration brings the condition number within a modest factor of its optimum over all diagonal scalings. We employ the iterative row/column scheme in the max norm, of the family analyzed by Ruiz [12]: six sweeps, each consisting of a row pass followed by a column pass,

$$\begin{aligned} s_i^{\text{row}} &= \max_j |\widetilde{M}_{ij}|, & \widetilde{M}_{i:} &\leftarrow \widetilde{M}_{i:}/s_i^{\text{row}}, & d_i^{\text{row}} &\leftarrow d_i^{\text{row}}/s_i^{\text{row}}, \\ s_j^{\text{col}} &= \max_i |\widetilde{M}_{ij}|, & \widetilde{M}_{:j} &\leftarrow \widetilde{M}_{:j}/s_j^{\text{col}}, & d_j^{\text{col}} &\leftarrow d_j^{\text{col}}/s_j^{\text{col}}, \end{aligned} \quad (18)$$

initialized with $\widetilde{\mathbf{M}} \leftarrow \mathbf{M}$ and $d_i^{\text{row}} = d_j^{\text{col}} = 1$. Only the diagonals are stored, as vectors Dr [27] and Dc [27]. After the sweeps,

$$\widetilde{\mathbf{M}} = \mathbf{D}_r \mathbf{M} \mathbf{D}_c, \quad \mathbf{D}_r = \text{diag}(d_0^{\text{row}}, \dots, d_{26}^{\text{row}}), \quad \mathbf{D}_c = \text{diag}(d_0^{\text{col}}, \dots, d_{26}^{\text{col}}), \quad (19)$$

with all diagonal entries positive and finite. Hence $\mathbf{D}_r$ and $\mathbf{D}_c$ are nonsingular and

$$\text{rank}(\widetilde{\mathbf{M}}) = \text{rank}(\mathbf{M}), \quad (20)$$

so equilibration alters the numerical scaling but not the exact rank. Consistency is preserved when the linear system is eventually solved: with $\widetilde{\mathbf{C}} = \mathbf{D}_r \mathbf{C}$, the solution $\mathbf{y}$ of $\widetilde{\mathbf{M}}\mathbf{y} = \widetilde{\mathbf{C}}$ yields $\Psi = \mathbf{D}_c \mathbf{y}$, since $\mathbf{D}_r \mathbf{M} \mathbf{D}_c \mathbf{y} = \mathbf{D}_r \mathbf{C}$ is equivalent to $\mathbf{M}\Psi = \mathbf{C}$.

3.3 Three-stage rank gates and acceptance threshold

Deterministic QRCP (Section 2) is applied in three stages. The first two stages test column-normalized copies of $\mathbf{X}$ and $\mathbf{W}$ individually; they serve as early-exit filters and as diagnostics that attribute a failure either to degenerate stencil geometry (rank loss in $\mathbf{X}$) or to an unsuitable radial-basis parameter pair (β, h) (rank loss in $\mathbf{W}$). They are necessary conditions for the admissibility of the product, because

$$\text{rank}(\mathbf{X}^\top \mathbf{W}) \leq \min\{\text{rank}(\mathbf{X}), \text{rank}(\mathbf{W})\}, \quad (21)$$

Algorithm 1 Numerical rank certification of a candidate triple (stencil, β , h)

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1: Assemble  $\mathbf{X}, \mathbf{W} \in \mathbb{R}^{27 \times 27}$ ; form  $\mathbf{M} = \mathbf{X}^\top \mathbf{W}$ 
2: Gate 1: QRCP on column-normalized  $\mathbf{X}$ ; if  $\text{rank}_{\text{num}}(\mathbf{X}) < 27$  reject (stencil geometry)
3: Gate 2: QRCP on column-normalized  $\mathbf{W}$ ; if  $\text{rank}_{\text{num}}(\mathbf{W}) < 27$  reject (RBF parameters)
4:  $\widetilde{\mathbf{M}} \leftarrow \mathbf{M}$ ;  $\text{Dr}[i] \leftarrow 1$ ,  $\text{Dc}[j] \leftarrow 1$ 
5: for  $k = 1, \dots, 6$  do ▷ equilibration sweeps, Eq. (18)
6:   row pass: divide each row of  $\widetilde{\mathbf{M}}$  by its max-norm; accumulate in  $\text{Dr}$ 
7:   column pass: divide each column of  $\widetilde{\mathbf{M}}$  by its max-norm; accumulate in  $\text{Dc}$ 
8: end for
9:  $\tau_{\widetilde{\mathbf{M}}} \leftarrow 27 \epsilon_{\text{dbl}} \|\widetilde{\mathbf{M}}\|_F$ 
10: Gate 3: deterministic QRCP on  $\widetilde{\mathbf{M}}^\top$ ; count pivots with  $p_i > \tau_{\widetilde{\mathbf{M}}}$ 
11: if count = 27 then
12:   accept; on demand solve  $\widetilde{\mathbf{M}} \mathbf{y} = \mathbf{D}_r \mathbf{C}$  and return  $\Psi = \mathbf{D}_c \mathbf{y}$ 
13: else
14:   reject; advance to the next (stencil,  $\beta$ ,  $h$ ) trial
15: end if

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but they are not sufficient: full rank of both factors does not imply full rank of their product. The decisive third stage therefore applies QRCP to $\widetilde{\mathbf{M}}^\top$ and counts as an independent direction every pivot whose residual magnitude—the trailing norm of Proposition 2.1, evaluated at the top of each step and compared against the tolerance frozen at entry (Proposition 2.3, Remark 2.4)—exceeds

$$p_i > \tau_{\widetilde{\mathbf{M}}}, \quad \tau_{\widetilde{\mathbf{M}}} = 27 \epsilon_{\text{dbl}} \|\widetilde{\mathbf{M}}\|_F. \quad (22)$$

The triple is accepted, and weights are exported, if and only if all three gates report numerical rank 27:

$$\text{rank}_{\text{num}}(\mathbf{X}) = 27, \quad \text{rank}_{\text{num}}(\mathbf{W}) = 27, \quad \text{rank}_{\text{num}}(\widetilde{\mathbf{M}}) = 27. \quad (23)$$

The complete procedure is summarized in Algorithm 1.

3.4 Justification of the threshold

The tolerance (22) is the member of the standard family (16) obtained with dimension factor $f = n = 27$ and the Frobenius norm as the computable measure of $\|\widetilde{\mathbf{M}}\|$. Its role follows directly from backward stability: the computed factorization corresponds exactly to a perturbed matrix $\widetilde{\mathbf{M}} + \Delta$ with $\|\Delta\|_F \leq c(n) \epsilon_{\text{dbl}} \|\widetilde{\mathbf{M}}\|_F$ [5, Thm. 19.4], so any pivot residual below the level $n \epsilon_{\text{dbl}} \|\widetilde{\mathbf{M}}\|_F$ lies within the uncertainty introduced by the arithmetic itself and cannot be distinguished from a numerically zero direction. Two remarks make the choice precise.

Remark 3.1 (Conservativeness of the Frobenius norm). Since $\|\mathbf{A}\|_2 \leq \|\mathbf{A}\|_F \leq \sqrt{n} \|\mathbf{A}\|_2$, the tolerance $27 \epsilon_{\text{dbl}} \|\widetilde{\mathbf{M}}\|_F$ is at least as large as the spectral-norm variant $27 \epsilon_{\text{dbl}} \sigma_{\max}(\widetilde{\mathbf{M}})$ recommended in [3, §5.4.1]. The test is therefore conservative—biased toward rejection—while avoiding the cost of estimating $\sigma_{\max}$. After the equilibration of Section 3.2 the entries of $\widetilde{\mathbf{M}}$ are of order unity, so the two norms differ by at most the benign factor $\sqrt{27}$.

Remark 3.2 (Nature of the certificate). A passed Gate 3 asserts that all 27 QRCP pivots of $\widetilde{\mathbf{M}}^\top$ exceed the roundoff-level tolerance (22) in IEEE double precision. In view of the Kahan counterexample [8] and the strong-RRQR theory of Gu and Eisenstat [7], this constitutes a *double-precision numerical certificate* of $\text{rank}_{\text{num}}(\mathbf{M}) = 27$ —sufficient for the reliable solution of $\mathbf{M}\Psi = \mathbf{C}$ in the same arithmetic—and not a symbolic proof of $\det(\mathbf{M}) \neq 0$. For equilibrated matrices of dimension 27 arising from non-adversarial stencil geometries, the gap between the

two notions is immaterial in practice, and QRCP-based rank determination is the established inexpensive surrogate for the singular value decomposition [6, 10].

Concluding note

The certification procedure of Section 3 is built entirely from classical, well-audited components: Householder reflections [1, 3, 4], the Businger–Golub column-pivoted QR factorization [2, 9, 10], rank-revealing interpretation and its known limitations [6, 8, 7], iterative max-norm equilibration [11, 12], and the standard backward-error tolerance convention [5, 3]. The only problem-specific ingredients are the dimension $n = 27$, fixed by the D3Q27 interpolation stencil, and the three-gate ordering, which converts the mathematical characterization—unique solvability if and only if $\text{rank}(\mathbf{M}) = 27$ —into an efficient and diagnosable selection loop over candidate triples (stencil, β , h).

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